

DENVER INTL, CO, USA

WMO: 725650

Lat: 39.833N

Lon: 104.658W

Elev: 5414

StdP: 12.04

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 03017

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.2	5.8	-7.7	4.4	11.8	-3.3	5.6	20.2	29.7	42.3	26.3	36.9	7.6	230	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	94.8	59.8	92.3	59.7	89.6	59.5	64.8	81.0	63.7	80.5	62.5	79.6	9.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	96.9	68.1	59.0	91.3	67.9	57.3	85.9	67.5	32.7	80.4	31.7	80.3	30.8	79.4	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.1	23.7	19.9	DB	-10.3	99.3	5.6	2.1	-14.4	100.8	-17.6	102.0	-20.8	103.1	-24.8	104.6	(4)
(5)			WB	-10.9	67.5	5.4	1.0	-14.8	68.2	-18.0	68.7	-21.0	69.3	-25.0	70.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	31.9	32.7	41.3	47.5	56.7	67.9	75.0	72.8	64.6	51.1	40.2	31.0		
	DBStd	18.14	11.11	11.55	10.65	9.47	8.82	7.36	5.22	5.15	8.47	10.29	11.06	11.45		
	HDD50	2579	565	489	301	154	39	1	0	0	9	111	320	590		
	HDD65	5874	1027	904	734	525	279	53	4	8	107	437	743	1053		
	CDD50	3009	3	4	33	80	248	539	775	707	446	145	27	2		
	CDD65	827	0	0	0	1	22	141	313	250	94	6	0	0		
	CDH74	10097	0	1	11	51	433	1941	3561	2662	1254	179	4	0		
	CDH80	4536	0	0	0	4	117	876	1820	1215	477	27	0	0		
(14) Wind	WSAvg	10.0	9.6	10.0	10.8	11.6	10.4	10.2	9.6	9.4	9.3	9.6	9.5	9.6		
(15) Precipitation	PrecAvg	15.3	0.4	0.5	0.8	1.9	2.3	1.9	2.1	1.9	1.5	1.0	0.5	0.5		
	PrecMax	21.6	0.9	1.4	2.7	5.6	4.5	4.0	5.1	4.4	7.5	2.5	2.2	2.1		
	PrecMin	7.9	0.0	0.0	0.0	0.2	0.6	0.4	0.3	0.2	0.0	0.2	0.1	0.0		
	PrecStd	3.4	0.2	0.3	0.6	1.1	1.2	1.1	1.4	1.1	1.5	0.6	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	67.7	75.3	79.1	88.3	97.1	98.6	95.9	93.1	83.2	73.6	64.3		
		MCWB	42.7	44.1	47.3	50.2	56.6	58.3	60.6	59.7	58.8	53.1	47.1	42.6		
	2%	DB	58.5	61.8	70.2	74.9	83.2	92.6	95.6	93.1	89.0	79.2	68.4	58.0		
		MCWB	40.3	41.7	45.6	48.7	54.8	58.5	60.4	59.6	57.2	51.5	45.5	40.1		
	5%	DB	53.2	56.6	65.4	70.9	79.0	89.2	93.1	90.3	85.7	74.8	63.5	53.1		
		MCWB	38.0	39.4	43.9	47.5	53.9	58.1	60.5	59.7	56.1	50.1	43.6	37.9		
	10%	DB	48.1	50.8	60.4	66.3	74.3	85.4	90.2	87.3	81.9	69.3	58.1	47.9		
		MCWB	35.7	36.6	42.0	46.3	52.9	58.0	60.6	59.9	55.7	48.4	41.2	35.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	45.0	48.9	53.2	60.7	65.5	66.5	66.0	63.4	55.8	48.4	43.5		
		MCDB	61.8	65.3	71.1	70.3	78.6	82.7	82.4	80.7	80.1	76.7	69.9	62.3		
	2%	WB	41.0	42.3	46.6	50.4	58.1	63.0	65.2	64.3	61.0	53.2	46.2	40.8		
		MCDB	56.8	60.4	67.1	68.8	74.4	79.9	81.2	80.6	77.8	72.3	66.8	57.3		
	5%	WB	38.4	39.8	44.6	48.8	56.1	61.3	64.1	63.2	59.2	51.3	44.1	38.3		
		MCDB	52.3	55.4	63.6	67.2	72.7	78.7	80.7	80.2	77.0	70.4	62.2	51.8		
	10%	WB	36.1	37.2	42.4	47.1	54.3	59.8	63.0	62.1	57.7	49.5	41.9	35.8		
		MCDB	47.6	50.1	58.8	64.5	70.5	78.3	80.2	78.8	75.3	67.1	57.1	47.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.5	23.9	25.9	25.4	25.1	27.9	27.5	27.0	27.3	25.8	25.1	23.3		
		MCDBR	28.9	31.8	32.6	32.8	31.9	33.2	31.1	30.5	31.6	32.7	31.7	29.3		
		MCWBR	15.7	16.6	15.4	14.3	11.9	10.6	8.7	8.6	10.2	13.3	15.2	16.2		
	5% WB	MCDBR	28.1	30.1	31.0	30.0	28.2	28.4	26.6	26.2	27.4	29.1	30.3	28.1		
		MCWBR	15.8	16.2	15.2	13.8	12.0	11.1	9.0	8.7	10.0	12.7	15.1	16.1		
(40) Clear-Sky Solar Irradiance	taub	0.237	0.245	0.265	0.296	0.315	0.324	0.351	0.342	0.306	0.267	0.249	0.239			
	taud	2.623	2.614	2.560	2.483	2.489	2.479	2.441	2.457	2.536	2.633	2.603	2.608			
	Ebn at Noon	307	318	318	311	305	301	292	292	299	304	298	298			
	Edh at Noon	21	24	29	33	34	34	36	34	29	24	21	20			
(44) All-Sky Solar Radiation	RadAvg	803	1108	1499	1797	2005	2291	2204	1967	1677	1224	878	704			
	RadStd	35	62	109	124	158	105	93	58	82	75	48	40			

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	N/A
(47) Regional (28 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A

Nomenclature: See separate page